MIL-A-21577/7B(SHIPS)
15 August 1968
SUPERSEDING
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(See 6.2)

MILITARY SPECIFICATION

AMPLIFIER-CONTROL GROUP AN/SIA-119

1. SCOPE

1.1 This specification covers the detailed requirements for Audio Amplifier Control Group AN/SIA-119.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-62 - Capacitors, Fixed, Electrolytic (D.C. Aluminum Electrolytic Polarized) General Specification for.
MIL-I-983 - Interior Communication Equipment, Naval Shipboard, Basic Design Requirements for.
MIL-C-3965 - Capacitors, Fixed, Non-solid Electrolytic (Tantalum Foil and Sintered-Slug), General Specifications for.
MIL-A-21577 - Amplifiers, Audio Frequency; and Amplifier-Control Groups - Shipboard Announcing.

STANDARDS

MILITARY

MIL-STD-275 - Printed Wiring for Electronic Equipment.
MIL-STD-7C1 - Preferred and Guidance List of Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Amplifier-Control Group AN/SIA-119 shall be in accordance with the applicable paragraphs of MIL-A-21577 and as specified herein. Whenever a requirement of MIL-A-21577 conflicts with a requirement of this specification, the requirement of this specification shall govern.

3.2 Equipment complement shall consist of one microphone preamplifier assembly, six power amplifiers and one common power supply in a common bulkhead mounted enclosure. No amplifier-oscillator assemblies are required. One of the six power amplifiers is considered a spare.

3.3 Capacitors. - Tantalum electrolytic capacitors conforming to MIL-C-3965 shall be used in preamplifiers and power amplifiers. Plug-in type electrolytics in accordance with MIL-C-62 may be utilized in the power supply.

3.4 Semiconductor devices. - Semiconductor devices shall be selected from MIL-STD-7C1.

3.5 Wiring. - Wiring employed shall have insulation type T as shown in the applicable table of MIL-I-983.

3.6 Forced air cooling. - Forced air cooling is not permitted.

3.7 Printed Wiring. - Printed wiring, when employed, shall be in accordance with standoff type lead terminations specified in MIL-STD-275.

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3.8 Output power. - The rated output power for the individual power amplifier assemblies shall be 20 watts, minimum rating.

3.9 Input facilities. - Amplifier-Control Group AN/SIA-119 shall have input facilities for two local microphone jackboxes, type IC/MJB.

3.10 Output facilities. - Facilities shall be provided for transferring the associated loudspeaker load from the output of a remote amplifier to the output of the AN/SIA-119 amplifier. This transfer shall occur when the press-to-talk push button on either of the two local microphones is depressed. When amplifier AN/SIA-119 is used as a booster amplifier, manual provisions shall be included to transfer the local loudspeaker groups from its own output to the output of the remote amplifier upon failure of local power supply or plate voltage supply.

3.11 Input circuit. - Additional facilities shall be provided in the control rack for attenuating a 70-volt signal transmitted from a remote amplifier for injection into the input of the amplifier assembly. This attenuator shall be resistive, and shall appear as a load not exceeding 1 watt.

3.12 Alarm signal priorities. - When used as a booster or emergency amplifier, with local microphone input facilities, alarm signals transmitted from the remote amplifier shall have priority over all features of the amplifier assembly.

3.13 Speech priorities. - When used as an emergency amplifier, either of the local microphones shall have priority over all transmissions except alarm signals arriving from the remote amplifier.

3.14 Microphone press-to-talk switches. - The operation of a microphone press-to-talk switch at either of the two local microphone stations shall:

(a) Connect the microphone audio circuit at that station to the input of the preamplifier assembly.
(b) Energize the busy indicator lamp at the other microphone station.
(c) Place the amplifier channel in the "ready" condition.
(d) Disconnect the adjacent loudspeaker circuit associated with that microphone circuit.

3.15 Busy indicator circuits. - Busy indicator circuits shall be provided for the two local microphone stations so that when either microphone station is in use the busy indicator lamp of the other station will be energized.

3.16 Local loudspeaker cut-out circuit. - Local loudspeaker cut-out circuits shall be provided for each of the two local microphone stations.

3.17 Power amplifier assembly outputs. - Each individual power amplifier shall have its output brought to separate output terminals.

3.18 Frequency response. - Frequency response at the power amplifiers shall be measured as specified in MIL-A-21577 except that the amplifier shall be delivering 10db below rated output at 1000 cycles per second.

3.19 Distortion. - Distortion of the power amplifier shall be measured as specified in MIL-A-21577, except that at 200 and 6000 cycles, the measurement shall be made at 6db below rated output.

3.20 Module mounting. - All modules shall be mounted in a vertical position to facilitate the unrestricted flow of air.

3.21 Test facilities. - A common voltmeter with switching facilities shall be provided on the control rack to determine proper power supply voltage for amplifiers and preamplifiers. The meter calibration and circuit component values shall be such that operating condition of amplifiers can be determined directly from the meter without reference to tables and charts.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling, examination and tests shall be as specified in MIL-A-21577.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be as specified in MIL-A-21577.

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6. NOTES

6.1 The notes of MIL-A-21577 are applicable to this document.

6.2 CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE TO THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

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